

1-Web Testing Types

A→1-Functional/Regression Testing

2. Load and Performance Testing

3. Web Site Security Testing

4. Web Site Management Testing 5. Mobile Web/App Testing.

2-What is Selenium?

A→is a free (open-source) automated testing framework used to validate web applications.

3-what Selenium's Tool Suite?

A→

1. Selenium Integrated Development Environment (IDE)

2. Selenium Remote Control (RC)

3. WebDriver

4. Selenium Grid.

4-what is Selenium Features?

A→

- 1) Selenium is an open source
- 2) simple-to-understand and easy-to-use interface.
- 3) Selenese is the language used to write Selenium Commands.

5-Why to choose Selenium IDE?

A→To create simple test cases and test suites and To test a web application against Firefox and Chrome only.

6-components Selenium IDE?

A→1. Menu Bar

2. Address Bar

3. Tool Bar

4. Test Case Pane.

7- what is Selenese?

A→A command tells Selenium what to do.

8-Benefits of Using Selenium IDE?

A→ Gives developers greater flexibility in executing the test cases.

9-Limitations of Selenium IDE?

A→

1) Not suitable for testing extensive data like Big Data

2) Cannot handle the dynamic part of web-based application

3) Connections with the database can not be tested.

10-What is Selenium RC?

A→was the main Selenium project that sustained for a long time before Selenium.

11-How WebDriver took over Selenium RC?

A→Selenium WebDriver's architecture is simpler than that of Selenium RC.

12-Pros of Selenium WebDriver?

A→

1)simpler installation than selenium RC 2)communication directly to the browser

13- what is cookie?

A→ A cookie is a small piece of data from a web site that is stored on a visitor's browser to help the website track the visitor's activity

14- what is session?

A→ a way of maintaining state information about a user's interactions with a website or web application.

15-what is IDOR

A→ stands for Insecure Direct Object References,, simply changes a parameter value that directly refers to a system object to another object the user isn't authorized for

16-Avoiding Insecure Direct Object References?

A→

1) Eliminate the direct object reference

2) Validate the direct object reference

17-Write 5 CMM levels?

A→ Level 1: Initial • Level 2: Repeatable • Level 3: Defined • Level 4: Quantitatively Managed • Level 5: Optimizing

18-Explain the difference between error and defect?

A→ Error: product that is uncovered before the s/w is delivered to the end-user

Defect: A flaw that is uncovered after delivery to the end-user

19- Explain Buffer Overflow?

A→ Buffer Overflow occurs when data is written into a buffer (like array) past its end.

20- Explain SQL Injection?

A→ SQL Injection is a technique for injecting SQL commands into user

21- Explain OS Command Injection?

A→ OS Command Injection arises when user-specified input is directly handed over to the operating system

22- Explain Integer Overflow or Wraparound?

A→ Integer Overflow error occurs when you attempt to store a larger value into an integral type than will fit.

23- Explain Infinite loops?

A→ Occurs when loop termination condition is Never happening

24- What is Debugging?

A→ Debugging is the process of finding and fixing errors or bugs in the source code of any software.

25- How does the debugging process work?

A→

1) Error identification 2) Error analysis 3) Fix and validation 4) Debugging and testing

26- Explain the difference between syntax errors and semantic errors?

→ Syntax errors: It is the equivalent of a typo or spelling error in word processing

Semantic errors: the improper use of programming statements

27- What are common debugging strategies?

A→ 1) Incremental program development 2) Backtracking 3) Remote debugging 4) Logging

28- What is Breakpoints?

A→ Ability to stop execution based on certain criteria is key when debugging.

29-What is Conditional Breakpoints?

A→ Stop execution only when the set condition is true.

30- Explain Minification.

A→ is the process of removing all unnecessary characters from JavaScript

31- Benefits of Minifying JavaScript Code.

A→

1) Reduction of page load time

2) Lower script execution time

32- WHAT IS GARBAGE COLLECTION?

A→ Process of automatic freeing objects No longer referenced by the program

33- WHY GARBAGE COLLECTION?

A→ 1-Free unreferenced objects. 2- Combat heap fragmentation

34-Disadvantages GARBAGE COLLECTION?

A→ Extra Overhead , Less control over scheduling of CPU time

35- Explain Island of Isolation?

A→ a group of objects that reference each other

 **note→ Languages without GC C, C++, D, Objective-C, Rust (AND)**
Languages with GC Java, Go, C#, PHP, Python, Scala, Haskell 🧐

36- Explain Memory Leak?

A→ is a situation when there are objects present in the heap that are no longer used, but the garbage collector is unable to remove them from memory

37- Explain Software Testing?

A→ consists of the dynamic verification of the behavior of a program on a finite set of test cases

38- Explain the difference between Black Box Testing, White Box Testing, and Gray Box Testing.

A→ Black box testing: is testing that ignores the internal code of a system

White box testing: is testing that takes into account the full internal code of a system

Gray Box testing: partial interaction of internal code of system/component

39- Explain the difference between Unit Test and Integration Test?

A→

Unit testing: focuses on the individual modules of the application

- the first level of testing
- It is a white-box testing
- It focuses on module specification.

Integration testing: focuses on the combined modules of the application

- It is performed after Unit testing and before System testing
- It is a black-box testing
- It focuses on interface specification.

40- Explain Random Testing.

A→test inputs are picked purely randomly

41- Explain Specification-Based Testing.

A→Depending on how the program specifications are expressed

42- Explain PartitionTesting.

A→The program input domain is divided into sub-domains

43- Explain the difference between Verification and Validation .

A→ Verification: Are the algorithms coded correctly?

Validation: Does it meet user requirements?

44- Explain Positive Testing.

A→ the unit under test does what it is supposed to do

45- Explain Negative Testing.

A→ the software does not do that it is not supposed to do

46- Explain test coverage techniques.

A→

1. Statement Coverage 2. Decision Coverage 3. Branch Coverage 4. Condition Coverage

47- Explain Statement Coverage.

A→ execution of all the executable statements in the source code at least once

48- What is covered by Statement Coverage?

A→ 1. Unused Statements 2. Dead Code: Unreachable code 3. Unused Branches

49- Explain Decision Coverage.

A→ Decision coverage reports the true or false outcomes of each Boolean expression

50- Explain Branch Coverage

A→ branch coverage, every outcome from a code module is tested

51- Explain the difference between Alpha testing and Beta testing

A→

Alpha testing: Conducted at the developer's Environment by end users

Beta testing: Conducted at end-user Environment

52- Explain [TestFixture].

A→ marks a class that contains tests

53- Explain [Test].

A→ marks a specific method inside a class

54- Explain [TestFixtureSetUp].

A→ will be run once; before all other tests

55- Explain [TestFixtureTearDown].

A→ will be ran once; after all other tests have run

56- Explain [SetUp].

A→ ran before each of the tests

57- Explain [TearDown].

A→ ran after each of the tests are ran